

Data Encodings

Data payload encodings in the format are identified by a code (number). These codes are assigned by the FDSN. A list of valid codes are as follows:

Code	Description
0	Text, UTF-8 allowed, use ASCII for maximum portability, no structure defined
1	16-bit integer (two's complement), little endian byte order
3	32-bit integer (two's complement), little endian byte order
4	32-bit floats (IEEE float), little endian byte order
5	64-bit floats (IEEE double), little endian byte order
10	Steim-1 integer compression, big endian byte order
11	Steim-2 integer compression, big endian byte order
19	Steim-3 integer compression, big endian (not in common use in archives)
100	Opaque data - only for use in special scenarios, not intended for archiving

Overview and description of the Steim-1 and Steim-2 compression encodings may be found in the SEED 2.4 manual, Appendix B.

Retroactive future encodings

New data encodings may be added to the format in the future without incrementing the format version. There is no default encoding, readers must check the encoding value to determine if the encoding is supported.

Retired encoding values, not allowed in this specification

The following numeric codes were used in earlier miniSEED versions and should *not* be used for encodings defined in the future:

- 2:** 24-bit integers
- 12:** GEOSCOPE multiplexed format 24-bit integer
- 13:** GEOSCOPE multiplexed format 16-bit gain ranged, 3-bit exponent
- 14:** GEOSCOPE multiplexed format 16-bit gain ranged, 4-bit exponent

- 15:** US National Network compression
- 16:** CDSN 16-bit gain ranged
- 17:** Graefenberg 16-bit gain ranged
- 18:** IPG-Strasbourg 16-bit gain ranged
- 30:** SRO format
- 31:** HGLP format
- 32:** DWWSSN gain ranged format
- 33:** RSTN 16-bit gain ranged format